Text Mining 2026

Lecture 2: Linguistics and Natural Language Processing

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Linguistics

Subdiscipline	Medium or unit	Natural language module
phonetics, phonology	audio signals	Automatic Speech Recognition
morphology	words, word formation	Segmentizers, tokenizers, Part-Of-Speech taggers, lemmatizers, compound splitters, multiword detection
syntax	sentences, grammatical structure and function	Syntactic parsers, chunkers
semantics	sentence or word meaning, compositionality, quantification	Semantic parsers, word-sense-disambiguation
pragmatics	language use in context	Context and domain models
methods	introspection, behavorism, neuro-cognitive models, rule-systems, empirical (experimental & stochastic), mathematical models, classification models, distributional models	
resources	Lexicons, grammars, text/video/audio collections and annotations (interpretations)	

Forms in language

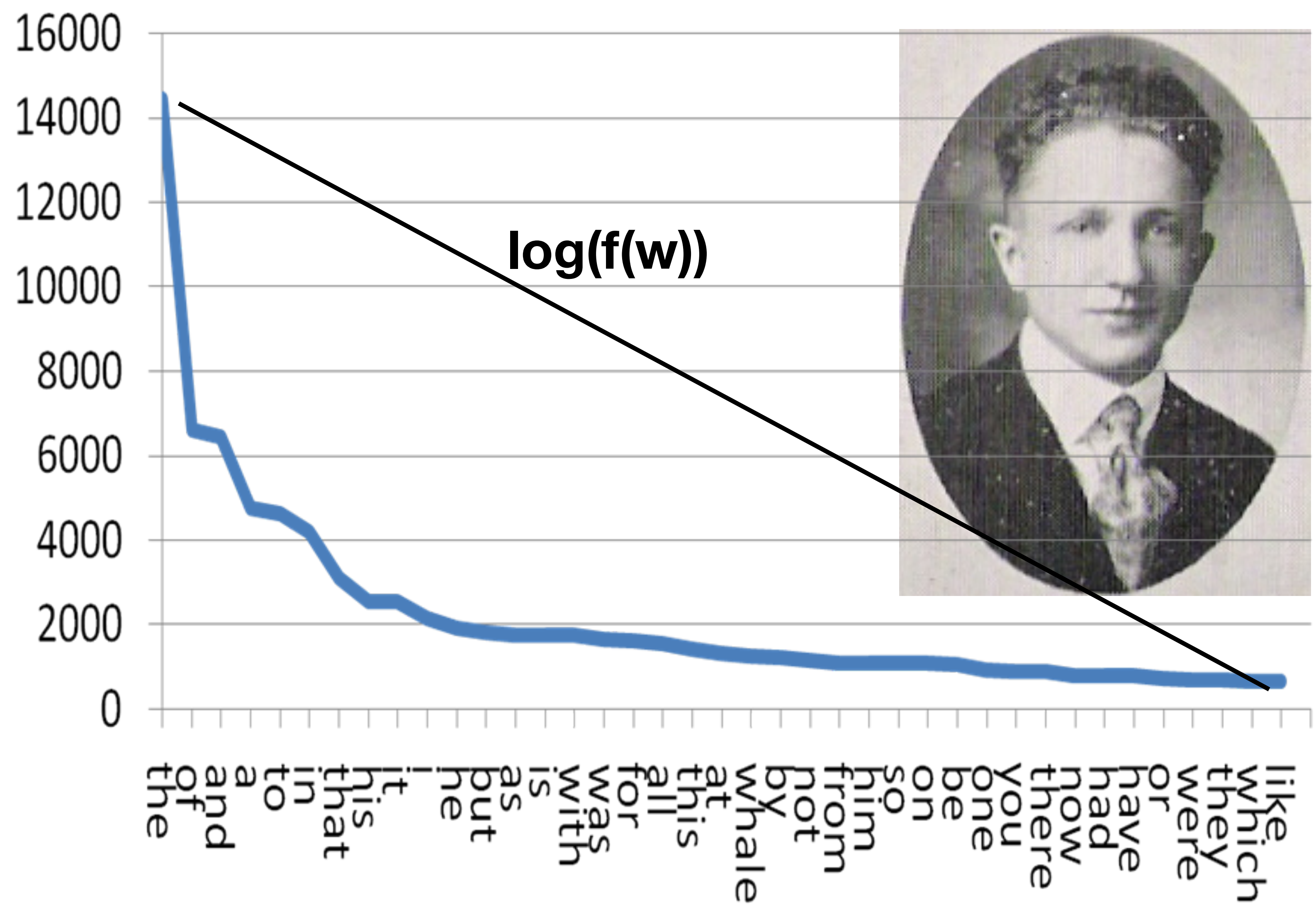
- A language has:
  - 11-112 phonemes (sound units), e.g. vowels “a”, consonants “k”
  - 4,000-10,000 morphemes (word units), e.g. “cat”, “un-“, “-ing”, “-ed”
  - 50,000 common words, millions of words including terminologies
  - An infinite number of sentences and expressions
- We use a small proportion of all these words very frequently (active knowledge) even though we recognise and understand many more (passive knowledge)
- Word, (Word) form, token —> used interchangeably in NLP

Zipfian distribution:

Power law distribution of word frequency

- $f(w_i)=f(W_1)/r_i(w_i)$
- Frequency of a word in a ranked list is the equal to the frequency of the most frequent word divided by the rank:
- Power law: 100, 50, 30, 25, 20, 16, 14, etc….
- Most frequent words tend to be short, have many different meanings and grammatical properties and make up 80% of any text.
- Words get meaning in context!

George Kingsley Zipf: 1902 - 1950



Morphology

- Study of the form and structure of words: what is a word, how many are there and what types?
- Words are composed of ***morphemes**, which are* the smallest meaning-bearing units:
    - e.g. "w*alked"* contains two morphemes: w*alk* (activity) and *-ed* (past);
    - "wa", "*alk*", "*lke*" are NOT morphemes in English
- **Free Morphemes**: occur independently, e.g. *boy, walk*
- **Bound Morphemes**: attached to another morpheme, cannot be used independently, and have some function, e.g. *-ed*
    - [INFLECTION]: modify a word
        - [TENSE past] -ed → walked
        - [NUMBER pl] -*s* → boys
    - [DERIVATION]: derive a word from another word (free form)
        - -ish, -ism, -ity, -ial —> boyish, racism, necessity, essential
    - **Affixes**: **prefixes** (e.g. *be*loved, *un*loved), **infixes** (e.g. minister-*s*-post (position of a minister), **suffixes** (e.g. love*able*)

Types of *words*

- Words can be classified by their **part-of-speech** (**PoS**)
  - specifies the typical phrase structures they "head" (see later)
- **Open class** (open to word formation and neologisms):
  - Noun (*boat, cow*), Verb (*float*), Adjective (*large, fast*), Adverb (*very, largely*)
  - new words easily invented and forgotten: "*chin diaper*"
- **Closed class** (you cannot easily introduce a new closed class word in a language):
  - Pronouns (*(s)he, him/her, it, this, who*), Prepositions (*in, at, from, in front of*).
  - relatively fixed, change slowly over generations (*gender diversity, "sher"*); small set of less than a hundred words
- **Stopwords are NOT a linguistic class of words**: frequent words with little content (open and closed class): *a, the, in, for, case, me, you, I, are, is, be, have, good*

The chin diaper is under the table.

My pronoun is sher.

Morphology

Word forms

- Word forms are all the inflected variants of a word: number, gender, case, tense, aspect, etc.
- Ratio forms to stems/roots: English 2:1, Dutch/German 5:1, Finnish/Turkish >200:1 (Crysmann 2006)
- Morphologically rich languages:
  - Finnish: 15 noun/adjective cases, 12 pronoun forms, 24 verb forms: Lentokonesuihkuturbiinimoottoriapumekaanikkoaliupseerioppilas (61),
    - = “an airplane jet turbine engine auxiliary mechanic non-commissioned officer student”
  - Turkish: *muvaffakiyetsizleştiricileştiriveremeyebileceklerimizdenmişsinizcesine* (71),
    - = "as if you are one of those that we cannot easily convert into an unsuccessful-person-maker”, https://en.wikipedia.org/wiki/Longest_word_in_Turkish
  - the number of ***unseen*** compounds in German newspapers grows linearly with the number of newspapers over time

Turkish verb forms

• gelmek - to come	• Gelemiyeceğini (söyledi) - (he said) he wouldn't be able to come
• gelmemek - not to come	• gelince - upon coming
• gelememek - not to be able to come	• geldiğimizde - when we came
• geliyorum - I am coming	• gelecekler - they will come
• geldim - I have come, I came	• gelmez - she/he does not come/won't come
• gelmişti - he had come	• Gel! - Come!
• gelirse - if he comes	• gelecektiniz - you will have come, you would come
• geliyorsa - if he is coming	• Geleceğine (ınanmıyorum) - (I don't believe) you will come
• geldiyse - if he came, if he had come	• geleceğimizi (söyledi) - (he said that) we will come, or (he said that) we would come.
ETC….	

Morphology

Part-of-Speech-tagging

- Task: assign the Part-of-Speech (PoS) category (e.g. noun, verb, adjective) to every token in a text

The	green	train	runs	down	that	track	.
Det	Adj (.8)/ NN(.2)	NS(.6)/ VI(.4)	NP(.3)/ VS(.7)	Prep(.4)/ Adv/Adj(.6)	Comp(.6)/ Pron(.4)	NS(.7)/ VI(.3)	.
Det	Adj	NS	VS	Prop	Pron	NS	.

- Collections of texts labeled by people with PoS tags
  - Choosing the prior already gives 90% accuracy
  - At least 50 different tag-sets used: http://universaldependencies.org/u/pos/
- Traditional PoS tagging uses Markov Models (sequence model):
  - the next state in the sequence is dependent on the current state

Morphology

Syntax

- Sequences of words exhibit structures and patterns
- We experience a sentences as a complete grammatical structure and have a strong intuition if a sentence is …
not complete.
- We can freely combine words into **phrases** or **constituents** and create an infinite number of sentences.

Phrases or constituents

- A **phrase** is a word or a group of words that functions as single unit within a grammatical structure
- A phrase is built around a ***head*** word with ***modifiers*** and can be nested**:**
  - ***very nice*** = Adjective Phrase or AP (head is an adjective (A), modifier is an adverb)
  - ***a very nice looping*** = NP (head is noun (N), modifier is an AP)
  - ***performs a very nice looping*** = VP (head is a verb (V), modifier is an NP)
  - ***with a long stick*** = Prepositional Phrase or PP (head is preposition (P))
  - ***the cow performs a very nice looping with a long stick*** = Sentence (S)

Phrase functions or dependency relations

- Dependency relations between heads of constituents:

The cow performs a very nice looping with a long stick

- main verb (perform), also the root: no outgoing arrows
- subject (cow), depends on main verb
- object (looping), depends on the main verb
- adjunct (stick), depends on the main verb
- modifier (nice), depends on looping
- modifier (long), depends on stick
- modifier (very), depends on nice

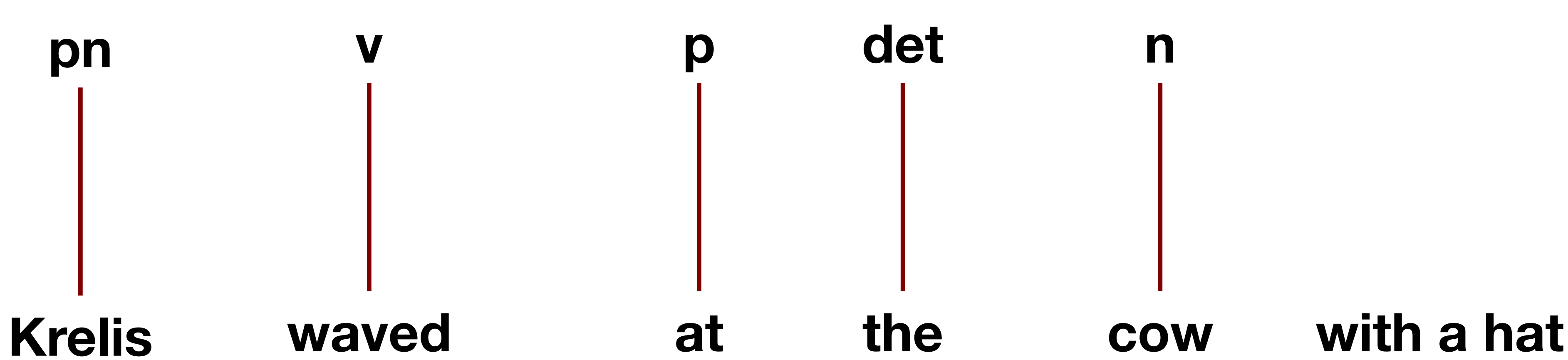
Syntax

Syntactic trees

Krelis waved at the cow with a hat

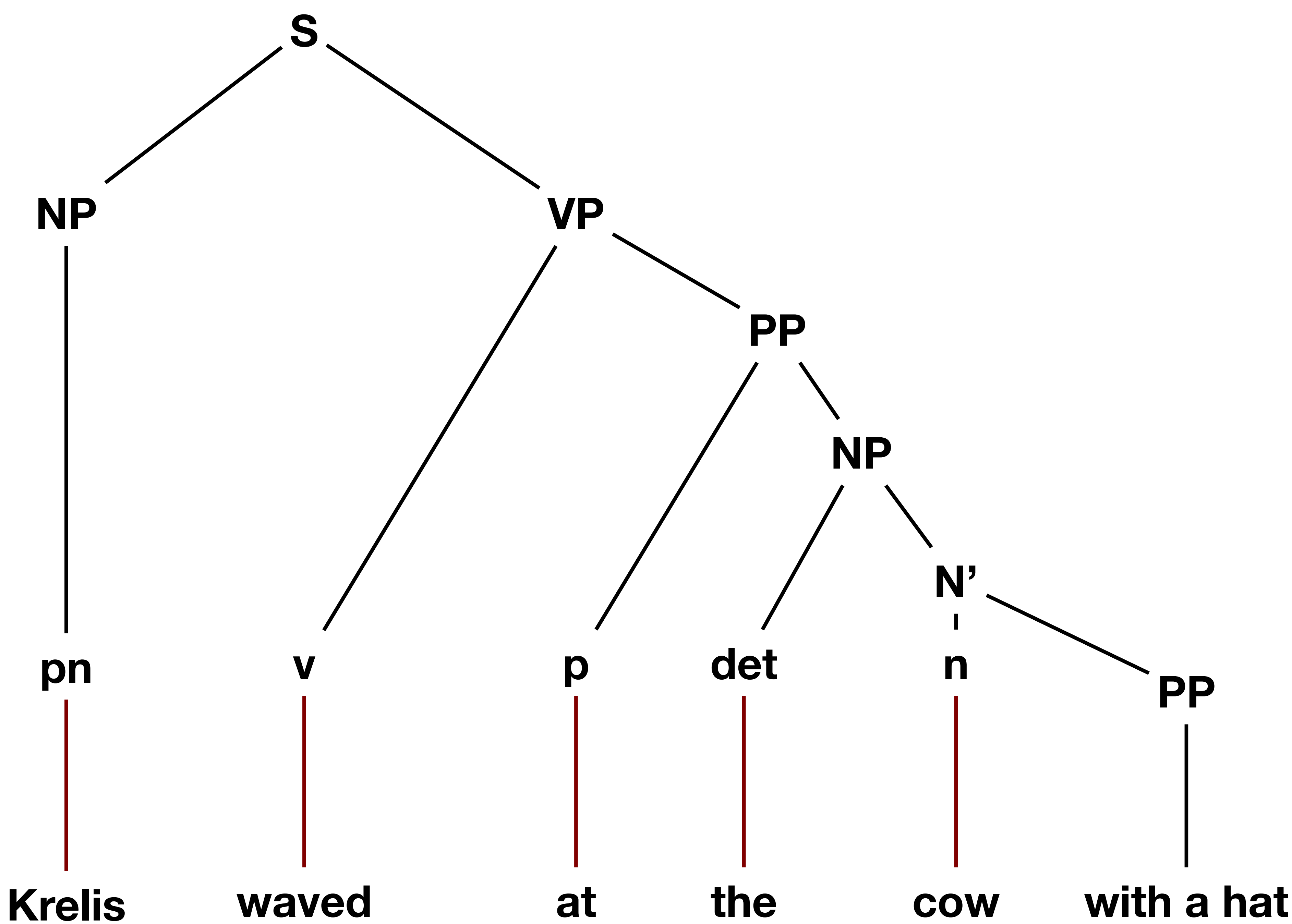
Syntax

Syntactic trees

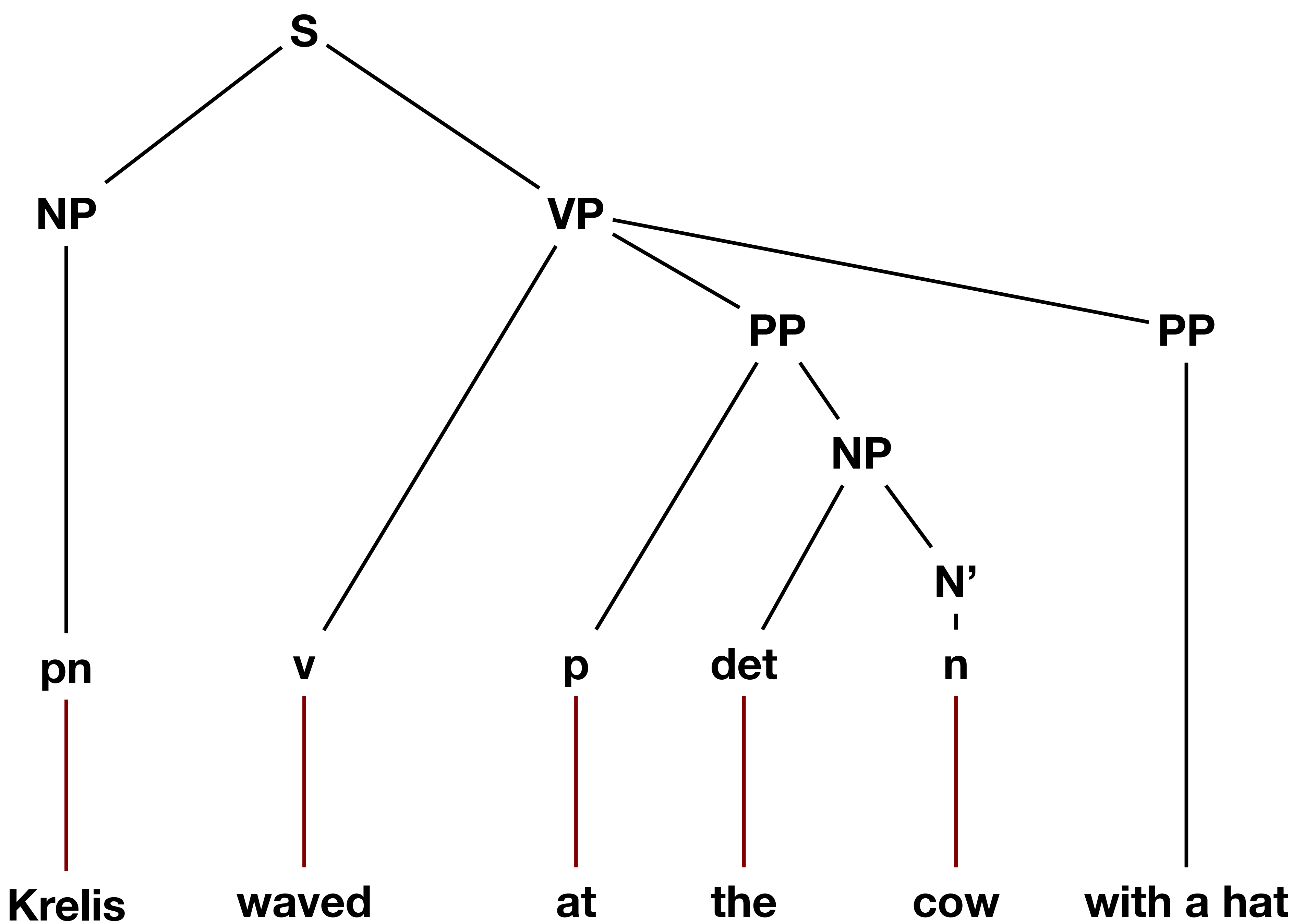


Syntax

Syntactic trees



Syntactic trees



Syntax

Syntactic functions

- Grammatical **Subject**: number agreement with main verb
  - **the *boys*$_{\textit{plural}}$ wave$_{\textit{plural}}$ at the cow$_{\textit{sing}}$; the *boy*$_{\textit{sing}}$ waves$_{\textit{sing}}$ at the cows$_{\textit{plural}}$**
  - **the *books*$_{\textit{plural}}$ were$_{\textit{plural}}$ given by me; the book$_{\textit{sing}}$ is$_{\textit{sing}}$ given by us**
  - **the *boys*$_{\textit{plural}}$ were$_{\textit{plural}}$ kicked by the cow; the *cow*$_{\textit{sing}}$ kicks the boys$_{\textit{plural}}$**
- Grammatical **Objects**: obligatory NPs or PPs that make a sentence grammatical in combination with a main verb
  - ***the teacher gives the students …?..**
  - ***the students fancy …?.**
  - ***the student treats the teachers …?.**
  - ***the cows perform …?.**
- https://universaldependencies.org/u/dep/

Syntax: some issues

- Constituents (S, VP, NP, PP, AP, AdvP) can be both very small and infinitely large
  - ***He (NP); The very nice black and white cows with red hats on their head (NP)***
- Language with little morphology (English) can be very ambiguous:
  - ***flies like sailing ships***
- PP-attachment is often context dependent
  - ***Krelis waved at the cow with a hat.***
- Scope is often context dependent
  - ***Old women and men can be annoying.***
  - ***Mary did not believe John murdered Bill with a knife.***
- Meaning is often context dependent
  - ***I count on my computer.***

Syntax: some issues

- Sentences contain typos and are often ungrammatical (esp. in Tweets and chat)!
    - ***My trainer don't tell nothing between rounds. All I want know I win round.***
    - ***It's out of the date and lacks validity, so formats of late exams are quite different from its. But for foreign language learners, 'there-insertion' is quite handful.***
- Conclusion: a parser should be robust.

If you don’t need full parse trees do ***chunking***

- Chunking (also called “shallow parsing”) provides a cheap and robust alternative to parsing
    - Chunks are like constituents
    - Chunkers do not provide full syntax trees, bit a list of constituents up to a certain depth (typically 2)
        - **[Krelis] [quickly waved] [at [the cow]] [with [a hat]]**
- After chunking another classifier can assign phrase types
    - **[Krelis]NP [quickly waved]VP [at [the cow]NP]PP [with [a hat]NP]PP**

Words have meanings

- The *head* of the department lost his *head* set after hitting his *head* against the *head* of the construction while *heading* for his meeting and enjoying the song *Head* by Prince.
  - https://en.wikipedia.org/wiki/Head_(disambiguation)

Semantics

WIKIPEDIA
The Free Encyclopedia

- Main page
- Contents
- Featured content
- Current events
- Random article
- Donate to Wikipedia
- Wikipedia store

Interaction

- Help
- About Wikipedia
- Community portal
- Recent changes
- Contact page

Tools

- What links here
- Related changes
- Upload file
- Special pages
- Permanent link
- Page information
- Wikidata item
- Cite this page

Print/export

- Create a book
- Download as PDF
- Printable version

Languages

- Deutsch
- Español
- Français
- 한국어
- Italiano
- Nederlands

Article Talk

Head (disambiguation)

From Wikipedia, the free encyclopedia

The **head** (**Human head**) is the part of an animal or human that usually includes the brain, eyes, ears,

Head may also refer to:

Arts, entertainment, and media [edit]

Music [edit]

Albums [edit]

- *Heads* (Bob James album), 1977
- *Head* (The Jesus Lizard album), 1990
- *Head* (the Monkees album), a 1968 soundtrack of the movie
- *Heads* (Osibisa album), 1972

Songs [edit]

- *Head* (The Cooper Temple Clause song), track from *Make This Your Own*
- "Head" (Julian Cope song), 1991
- "Head" (Prince song)
- "Head", a song by Mark Lanegan from *Bubblegum*
- "Head", a song by Static-X from *Beneath... Between... Beyond...*
- "Head", a song by Todd Sheaffer from *The Black Bear Sessions* and *Elko*
- "Head", a song by Lotion from *full Isaac*
- "Heads", a song by Hawkwind from *The Xenon Codex*

Other music [edit]

- Head (band), an English rock band
- The Head (band), an indie rock band from Atlanta, Georgia
- Head (music), a main theme in jazz
- Drumhead, a membrane on a drum
- Headstock, a part of an instrument

Film and television [edit]

- *Head* (film), a 1968 film starring The Monkees
- *Heads* (film), a 1994 TV movie
- *The Head* (film), a 1959 German horror film directed by Victor Trivas
- *The Head*, a 1994–1996 American animated television series
- "Head" (*Blackadder*), a 1986 episode of *Blackadder*
- Head (*American Horror Story*), a 2013 episode of the anthology television series

WordNet Search - 3.1
- WordNet home page - Glossary - Help

Word to search for: head [Search WordNet]

Display Options: (Select option to change) [Change]

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations
Display options for sense: (gloss) "an example sentence"
Display options for word: word#sense number

Noun

- S: (n) **head#1**, caput#2 (the upper part of the human body or the front part of the body in animals; contains the face and brains) *"he stuck his head out the window"*
- S: (n) **head#2** (a single domestic animal) *"200 head of cattle"*
- S: (n) mind#1, **head#3**, brain#3, psyche#1, nous#2 (that which is responsible for one's thoughts, feelings, and conscious brain functions; the seat of the faculty of reason) *"his mind wandered"; "I couldn't get his words out of my head"*
- S: (n) **head#4**, chief#1, top dog#1 (a person who is in charge) *"the head of the whole operation"*
- S: (n) **head#5** (the front of a military formation or procession) *"the head of the column advanced boldly"; "they were at the head of the attack"*
- S: (n) **head#6** (the pressure exerted by a fluid) *"a head of steam"*
- S: (n) **head#7** (the top of something) *"the head of the stairs"; "the head of the page"; "the head of the list"*
- S: (n) fountainhead#2, headspring#1, **head#8** (the source of water from which a stream arises) *"they tracked him back toward the head of the stream"*
- S: (n) **head#9**, head word#2 ((grammar) the word in a grammatical constituent that plays the same grammatical role as the whole constituent)
- S: (n) **head#10** (the tip of an abscess (where the pus accumulates))
- S: (n) **head#11** (the length or height based on the size of a human or animal head) *"he is two heads taller than his little sister"; "his horse won by a head"*
- S: (n) capitulum#1, **head#12** (a dense cluster of flowers or foliage) *"a head of cauliflower"; "a head of lettuce"*
- S: (n) principal#2, school principal#1, head teacher#1, **head#13** (the educator who has executive authority for a school) *"she sent unruly pupils to see the principal"*
- S: (n) **head#14** (an individual person) *"tickets are $5 per head"*
- S: (n) **head#15** (a user of (usually soft) drugs) *"the office was full of secret heads"*
- S: (n) promontory#1, headland#1, **head#16**, foreland#1 (a natural elevation (especially a rocky one that juts out into the sea))
- S: (n) **head#17** (a rounded compact mass) *"the head of a comet"*
- S: (n) **head#18** (the foam or froth that accumulates at the top when you pour an effervescent liquid into a container) *"the beer had a large head of foam"*
- S: (n) forefront#1, **head#19** (the part in the front or nearest the viewer) *"he was in the forefront"; "he was at the head of the column"*
- S: (n) pass#9, **head#20**, straits#2 (a difficult juncture) *"a pretty pass"; "matters came to a head yesterday"*
- S: (n) headway#2, **head#21** (forward movement) *"the ship made little headway against the gale"*
- S: (n) point#20, **head#22** (a V-shaped mark at one end of an arrow pointer) *"the point of the arrow was due north"*
- S: (n) question#2, **head#23** (the subject matter at *disease merits serious discussion"; "under the hea*
- S: (n) heading#1, header#1, **head#24** (a line of tex the passage below it is about) *"the heading seemed to have little to do with*

Semantics

Sentences have meanings
who did *what to whom, when, where, how*

- The man **opened** the door with a key
  - the man = NP$_{\text{[phrase]}}$, subject$_{\text{[dependency]}}$, **agent**$_{\text{[role]}}$
  - the door= NP$_{\text{[phrase]}}$, direct object$_{\text{[dependency]}}$, **patient**$_{\text{[role]}}$
  - with a key = PP$_{\text{[phrase]}}$, adjunct$_{\text{[dependency]}}$, **instrument**$_{\text{[role]}}$
- The door was **opened** by the man with the key
- The man stood before the door. The key **opened** the door
- The man took the key. The door **opened**
- For text mining you need to be able to handle different ways of “linguistic packaging” of the same content (!)
  - Different words, different syntactic structures express the same message

Semantic roles

- **Agent:** performs with control (can stop doing it)
- **Patient:** undergoes the action and is changed by it
- **Instrument:** what the agent uses to perform the action
- **Others:** recipient, thema, source, path, goal …

Semantic parsing

The boy ran from the shop across the street to his mummy

The boy fell

Harvey bought her flowers

She got flowers from Harvey

Flowers were given to her by Harvey

She broke Harveys eye socket

The hammer broke his eye socket

His eye socket broke

What colors correspond with what semantic relations?

Agent, patient, theme, beneficiary/recipient, instrument,

Source, path, goal,

Pragmatics

- In real live language is stretched to serve a purpose in a context (*people always try to make sense*)
  - ***Save your data up in the cloud.*** (**Metaphor**)
  - ***The three pizzas still need to pay.*** (**Metonymy**)
  - ***Can you close the window, please?*** (Form: question, Meaning: request)
  - ***It is a bit cold here, isn't it?*** (Form: a declarative sentence, Meaning: request)

cloud

mobile

Why do I need to know about linguistics?
We now have Large Language Models

Word embeddings

- Represent words as dense high-dimensional vectors, so-called word embeddings:
  - **actor** = [-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions
- Vectors are the weights of a neural network that learns from data:
  - to predict words that occur in a word's context (to the left or right) such as "*movie*", "*act*", "*play*"
  - and do not predict words that do not occur such as "*cat*", "*planet*", "*moon*".
- How?
  - Make the vector for actor (word embedding) more similar to the vector of words that do occur in the context and less similar to the vectors of words that do not occur in the context

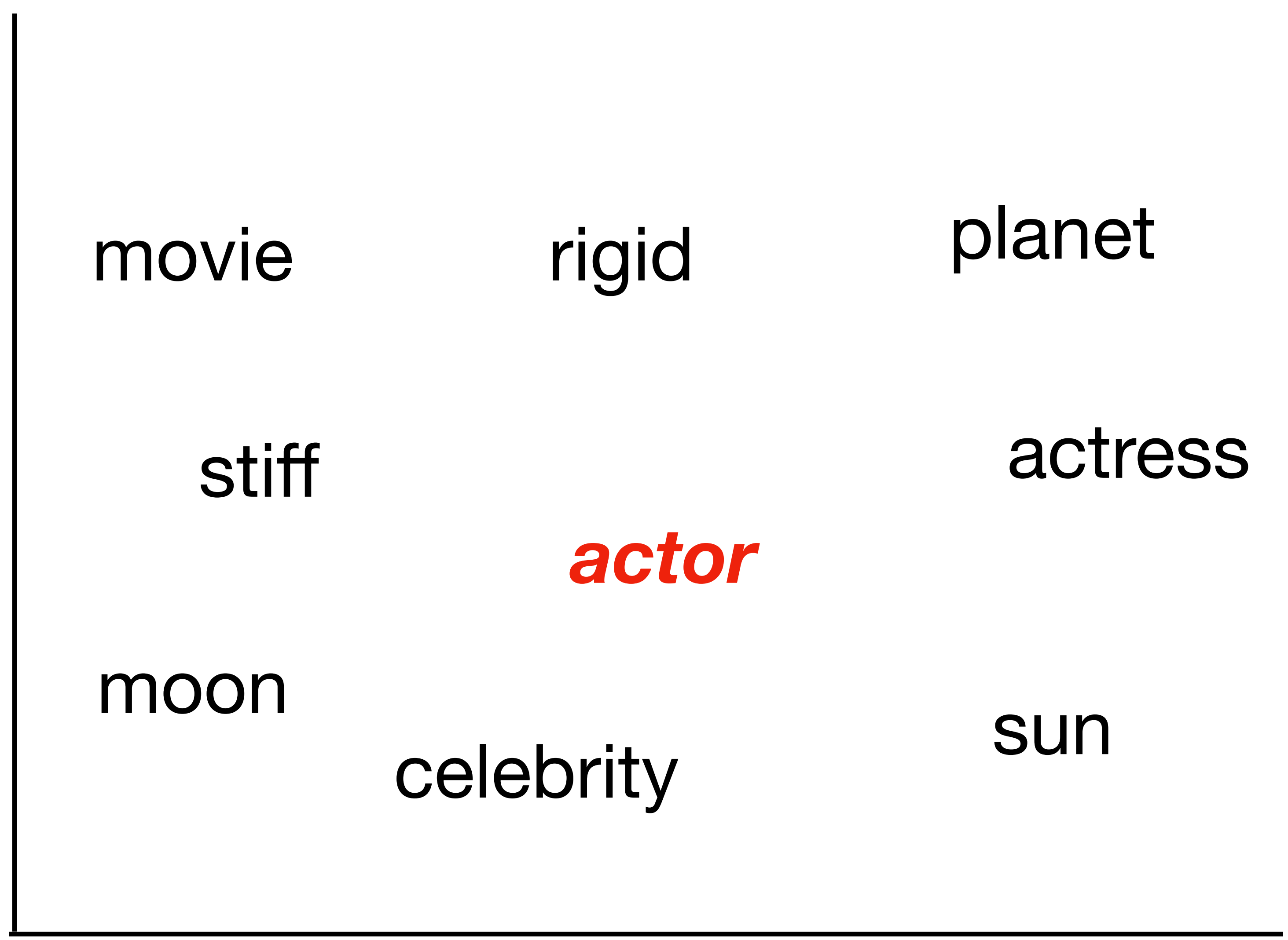
Word embeddings

actor [-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions

movie [-1.50961745E+00, -1.08213568E+00 2.62425601E-01, -1.23078614E-01, -1.27540553E+00, -5.51246941E-01, -3.492589E-01, -6.53275609E-01, …, etc, ……………..] 768 dimensions

Cosine Similarity

random initialisation at start



Word embeddings

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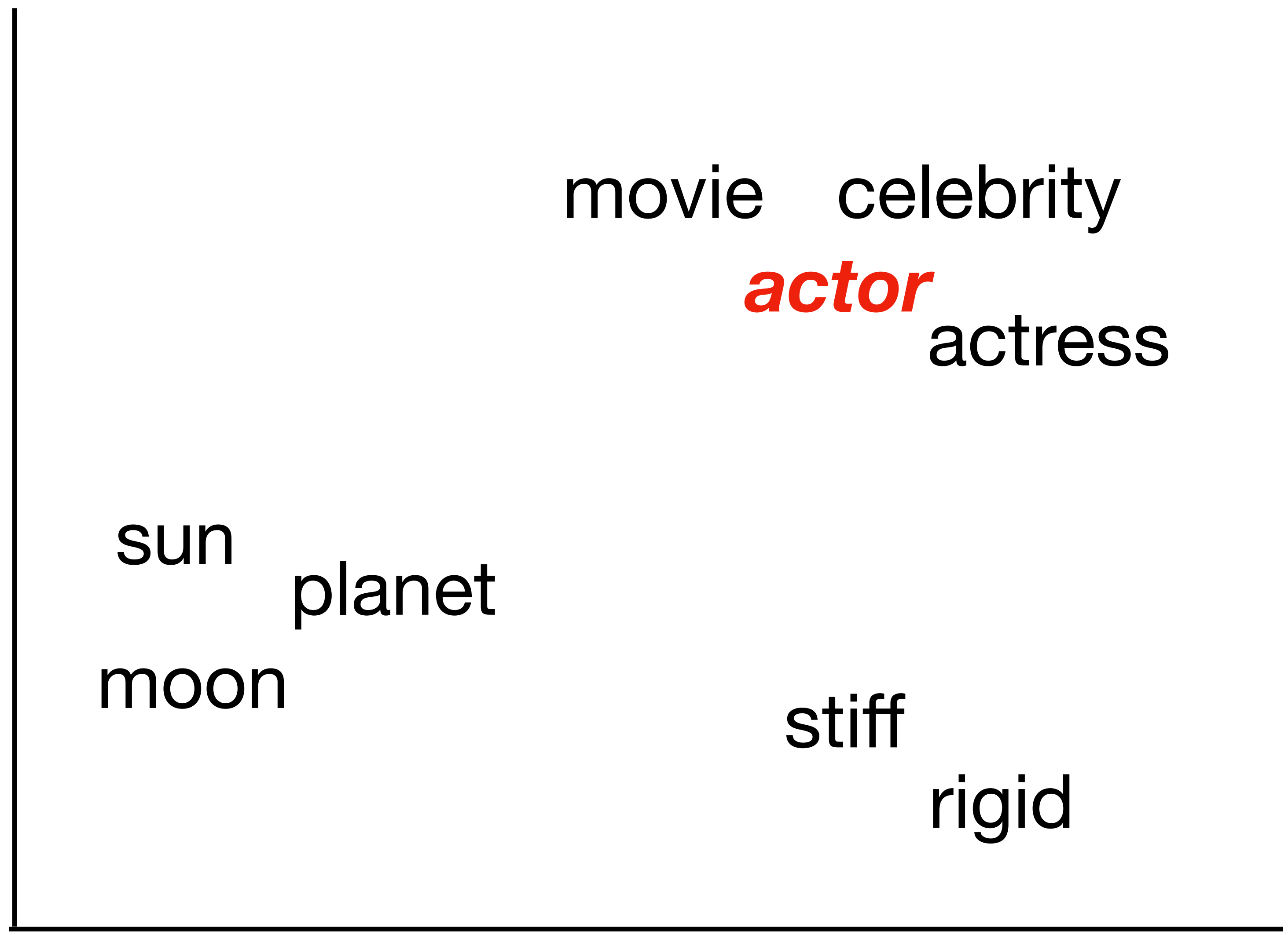
Cosine Similarity

converging after rounds of training



Word embeddings

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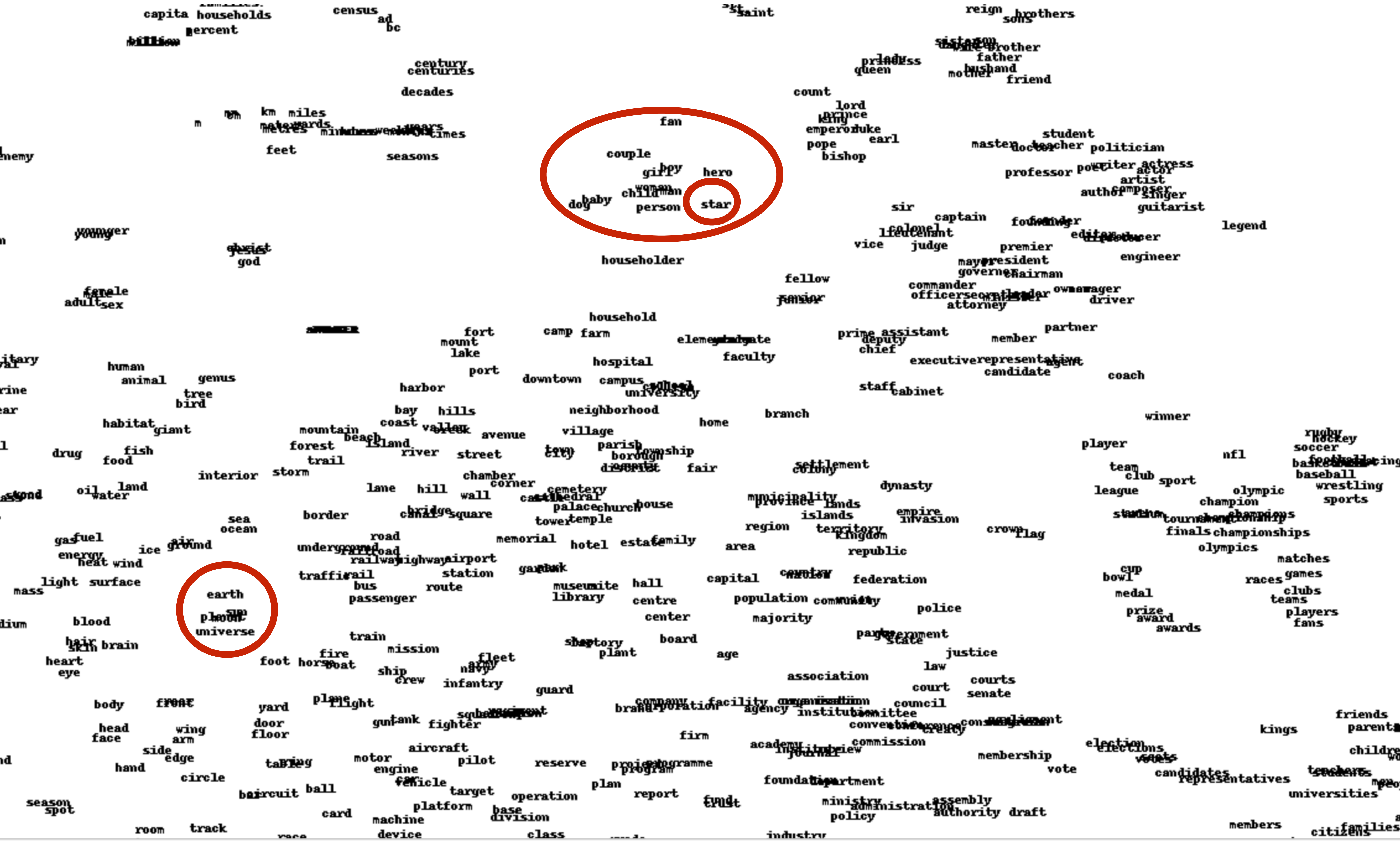
star [-1.50961745E+00, -1.08213568E+00 2.62425601E-01, -1.23078614E-01, -1.27540553E+00, -5.51246941E-01, -3.492589E-01, -6.53275609E-01, …, etc, ……………..] 768 dimensions

Cosine Similarity

The polysemy of stars



Joseph Turian's map of 2500 English words produced by using t-SNE on the word feature vectors learned by Collobert & Weston, ICML 2008

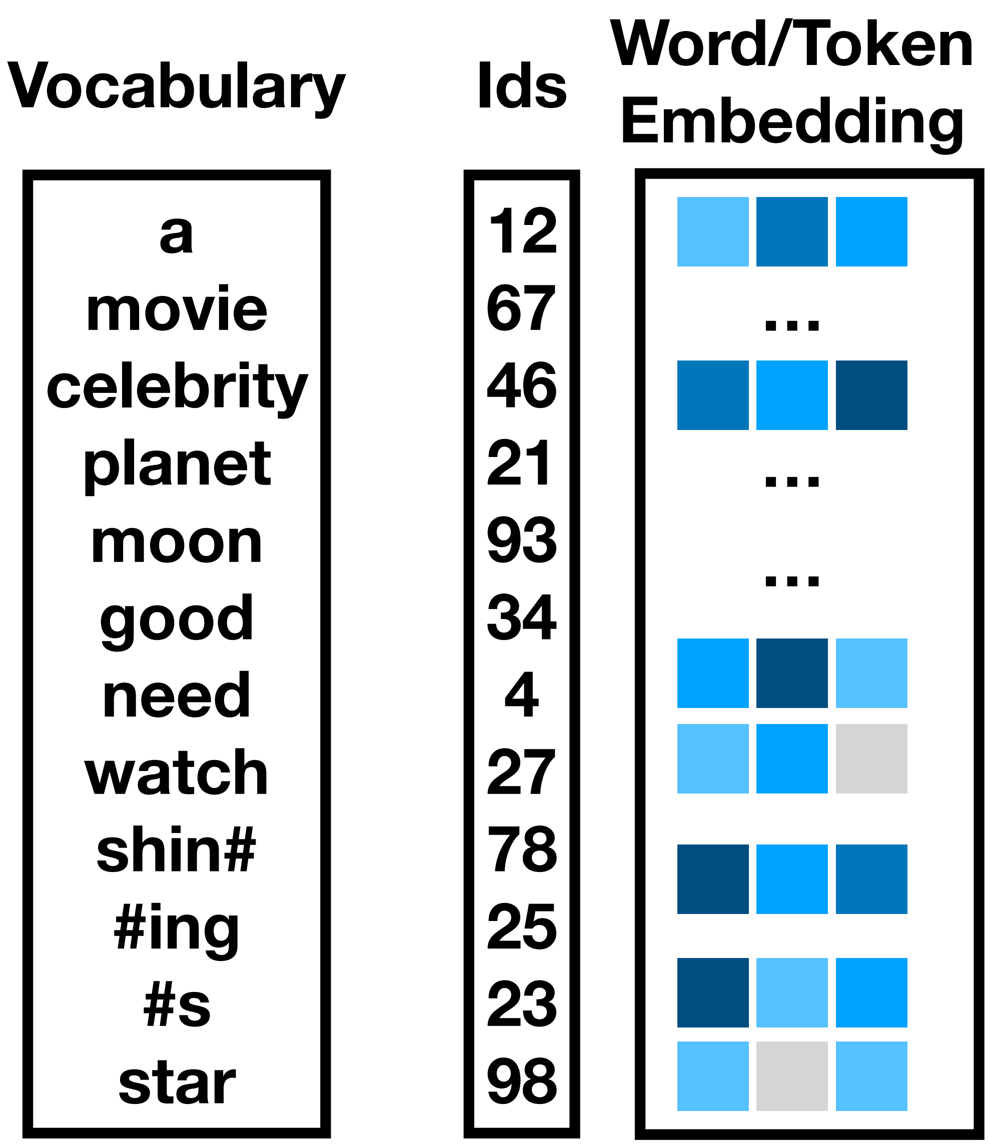
Large Language Models

- Represent words as dense high-dimensional vectors, so-called word embeddings:
  - star= [-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, …………….] 768 dimensions
- Learning objective: predict words ***in a specific context***, e.g. ***“A Hollywood star in a new […]”*** —> movie, show
- How?
  - Attention: combining the vector embedding of a word with the vector embedding of other tokens in the context to improve the predictions
- Result is a representation of sequences of tokens in context in which the weights are optimised *to predict each other with the help of the rest*

Stars in context

word/token embedding


[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, …………….] 768 dimensions



Stars in context

word/token embedding

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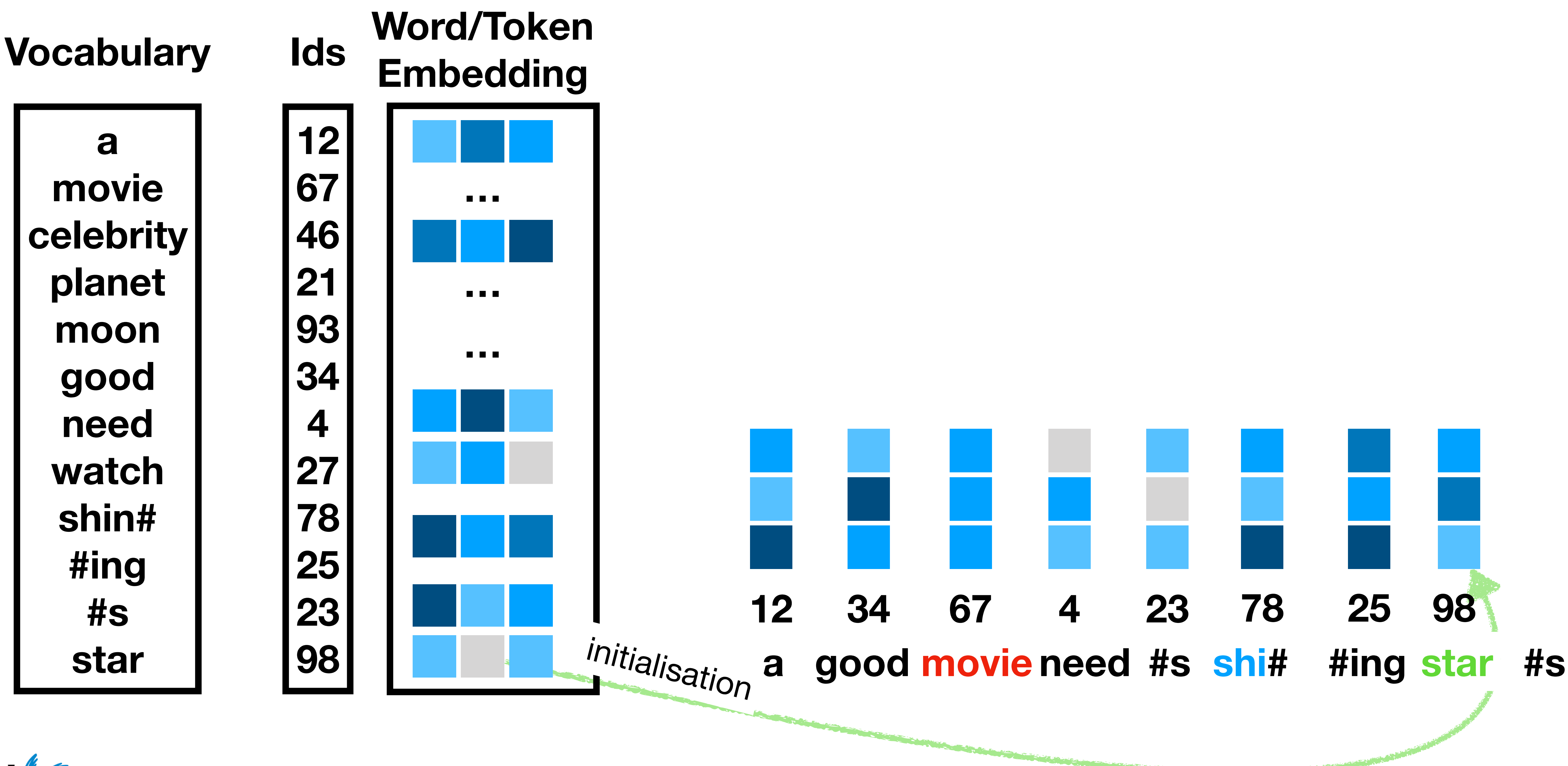
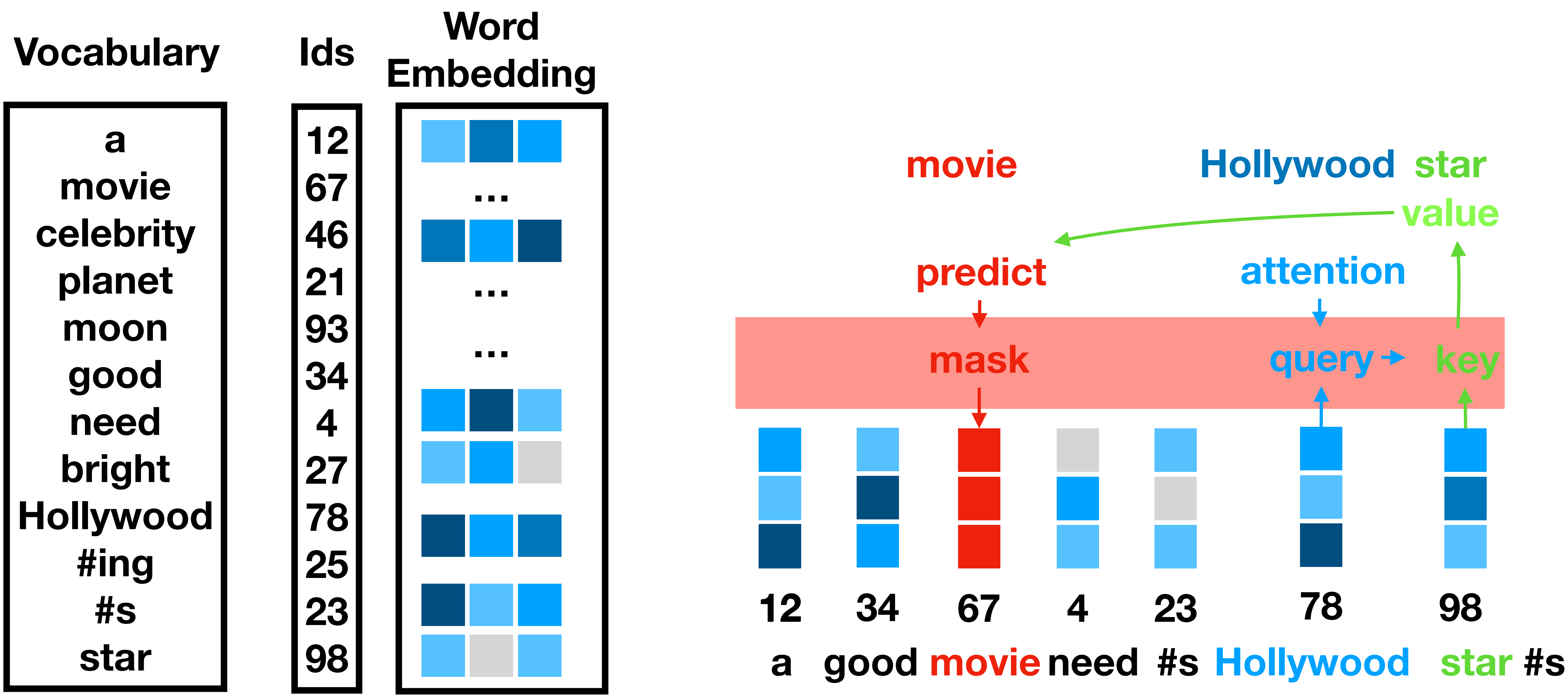


Stars in context



Attention

query = star [.1,.2,.3,.4]

X

key = Hollywood [.2,.2,.1,.6]

value = [mask] [.1,.2,.2,.5]

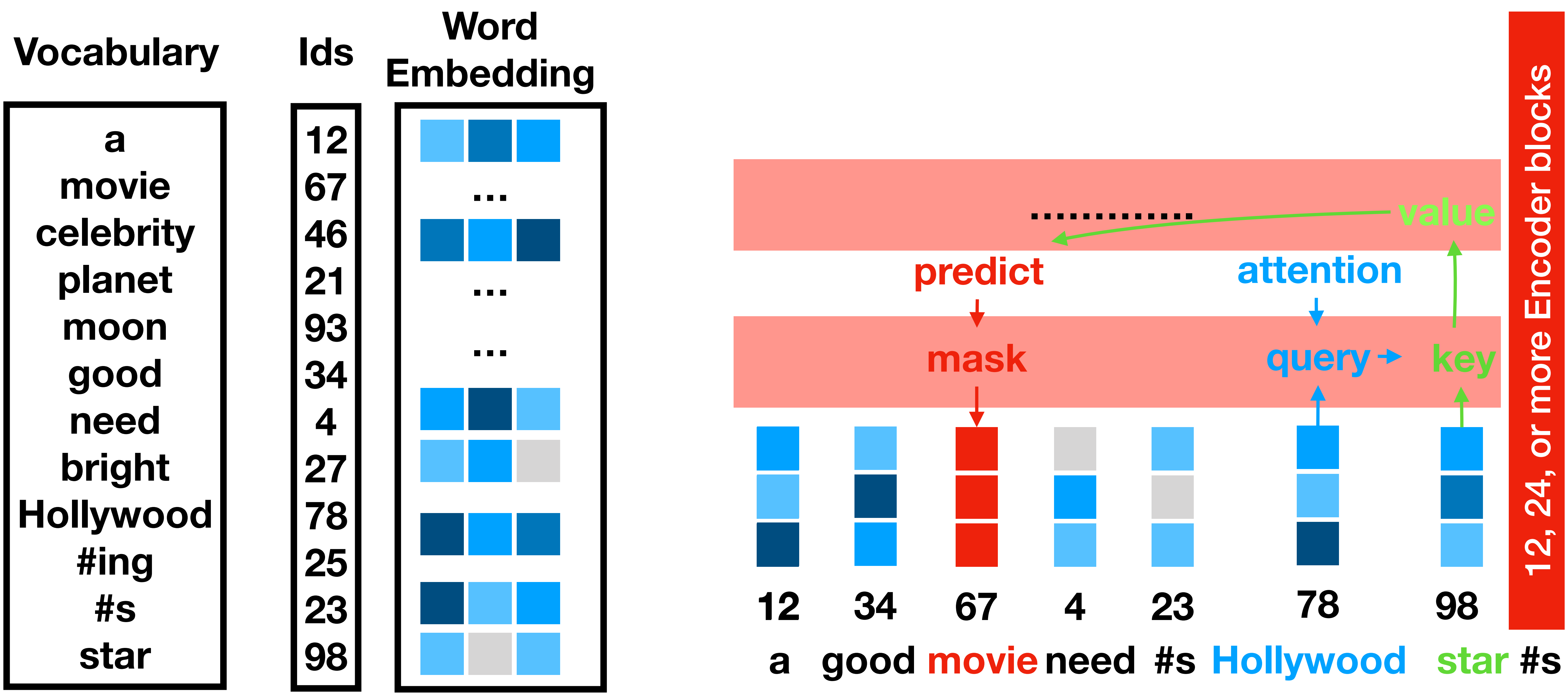
movie	[.1,.2,.2,.4]
show	[.1,.1,.2,.3]
series	[.2,.1,.2,.5]
planet	[.7,.3,.1,.2]
zebra	[.1,.8,.4,.1]

Loss function

Stars in context

word embedding

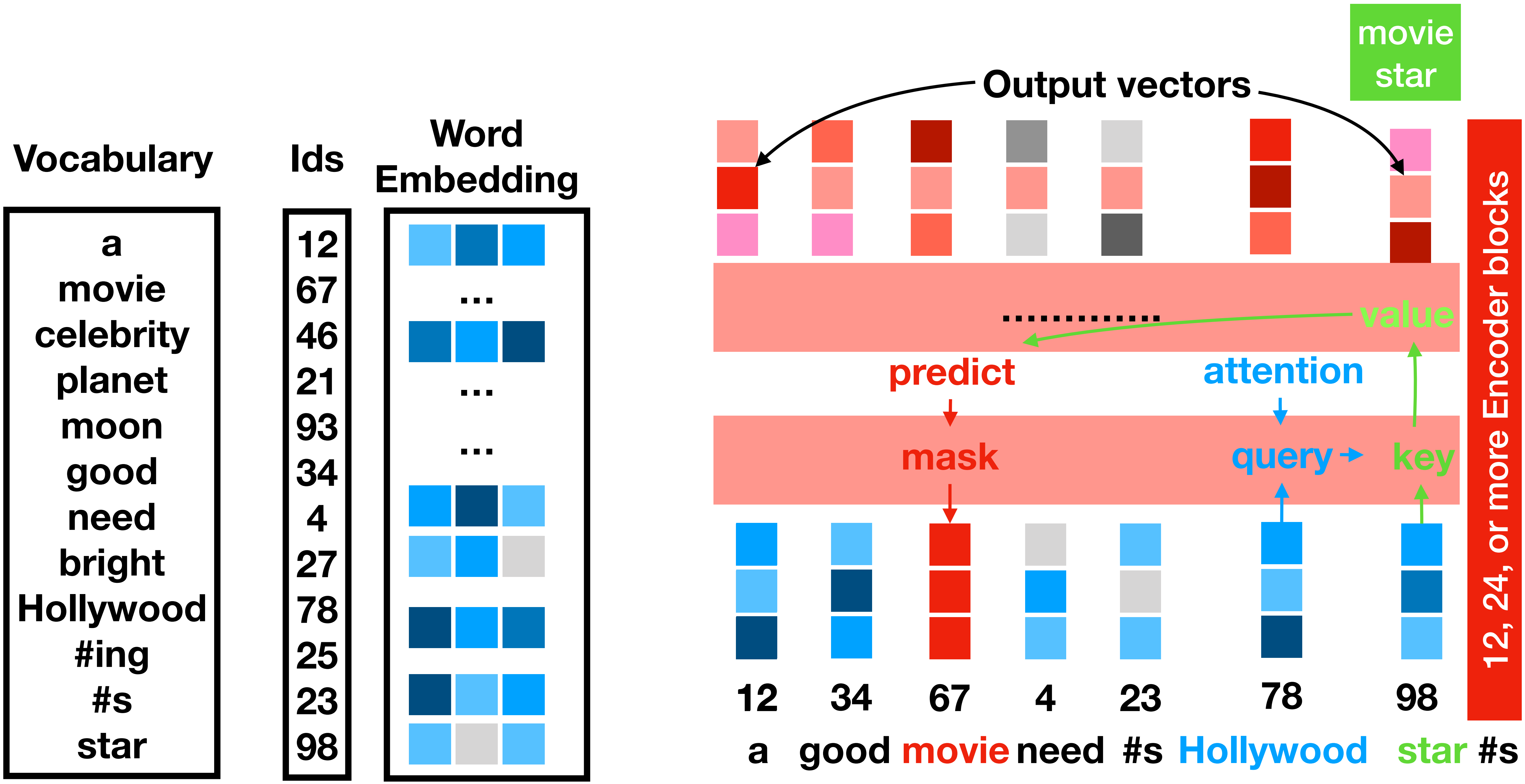
[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions



Stars in context

word embedding

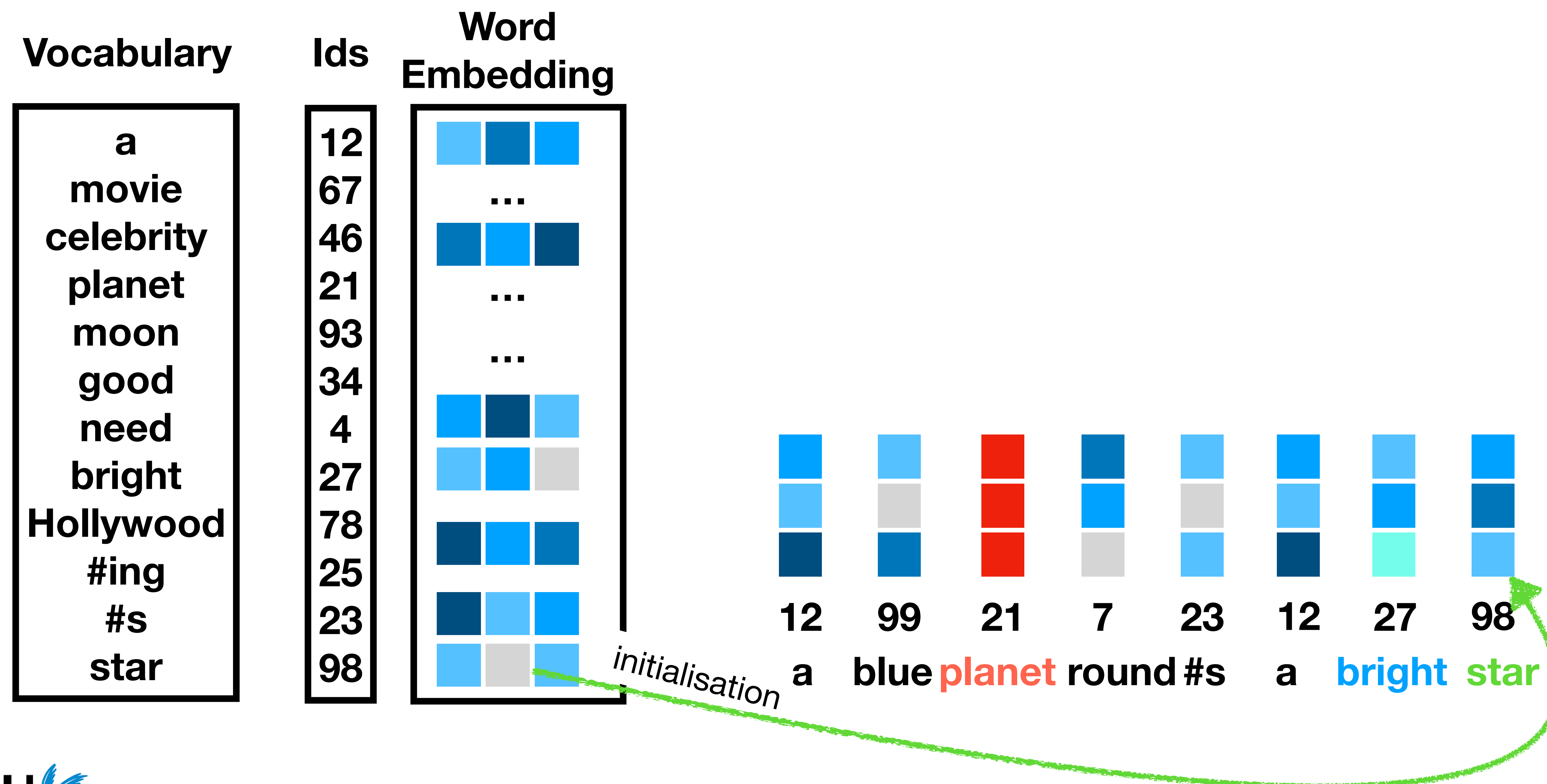
[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions



Stars in another context

word embedding

[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, ..., etc, ...] 768 dimensions

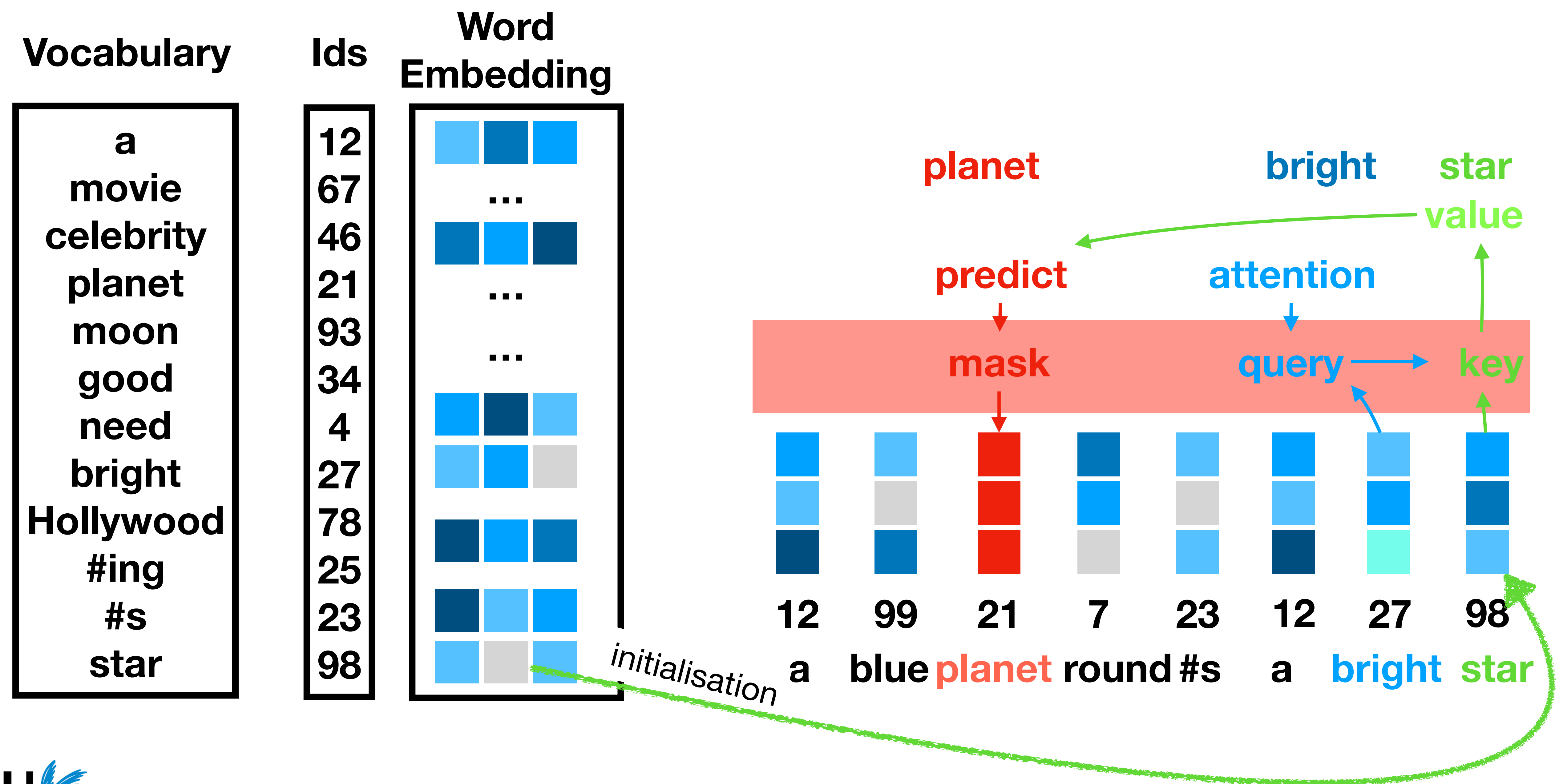


VU

Stars in another context

word embedding

[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions



Attention

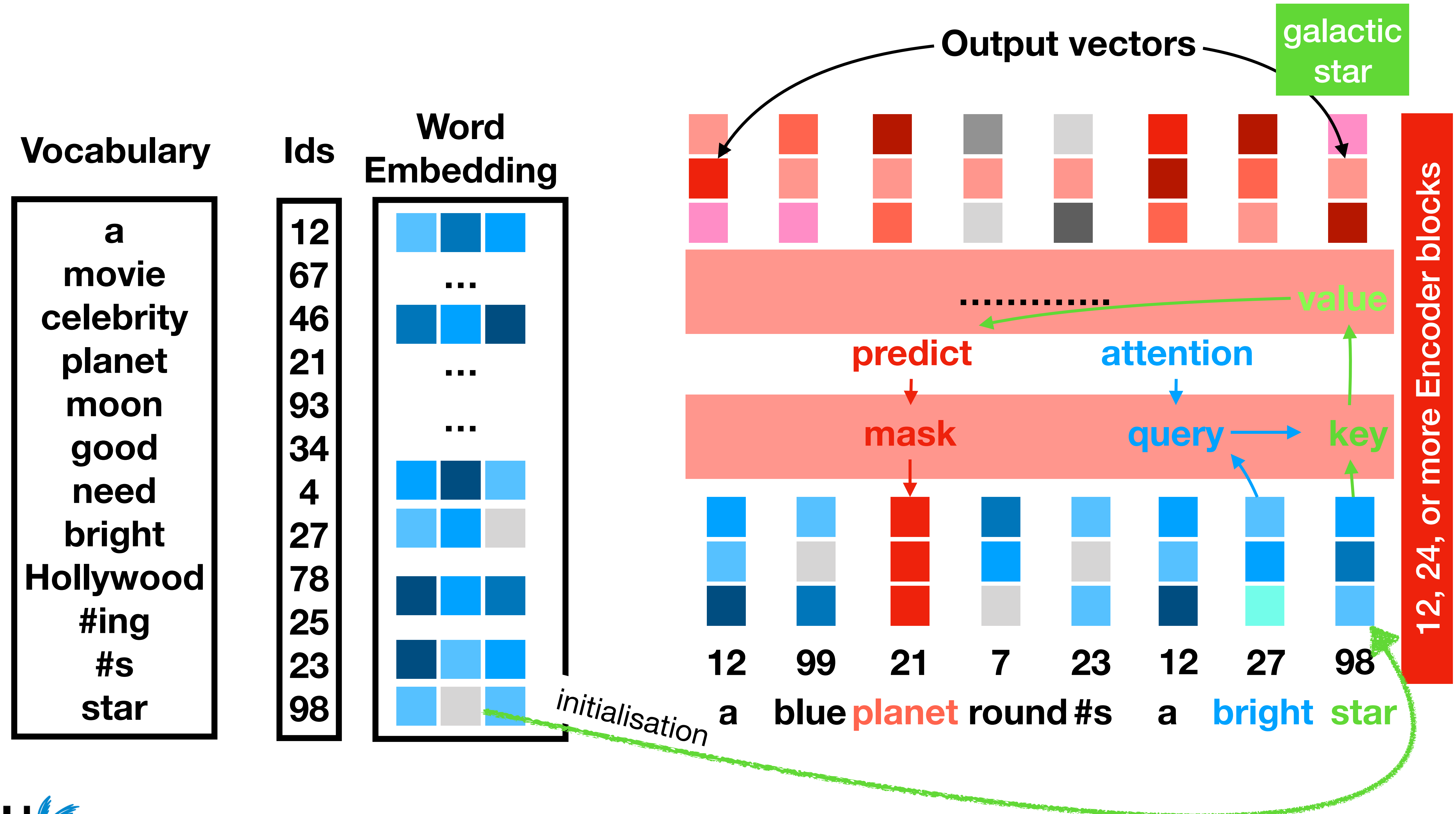
query = star	[.1,.2,.3,.4]	query = star	[.1,.2,.3,.4]
	X		X
key = Hollywood	[.2,.2,.1,.6]	key = bright	[.6,.3,.1,.1]
value = [mask]	[.1,.2,.2,.5]	value = [mask]	[.6,.3,.2,.2]
movie	[.1,.2,.2,.4]	planet	[.7,.3,.1,.2]
show	[.1,.1,.2,.3]	comet	[.5,.1,.2,.2]
series	[.2,.1,.2,.5]	astroid	[.6,.1,.1,.3]
planet	[.7,.3,.1,.2]	movie	[.1,.2,.2,.4]
zebra	[.1,.8,.4,.1]	zebra	[.1,.8,.4,.1]

Loss function **Loss function**

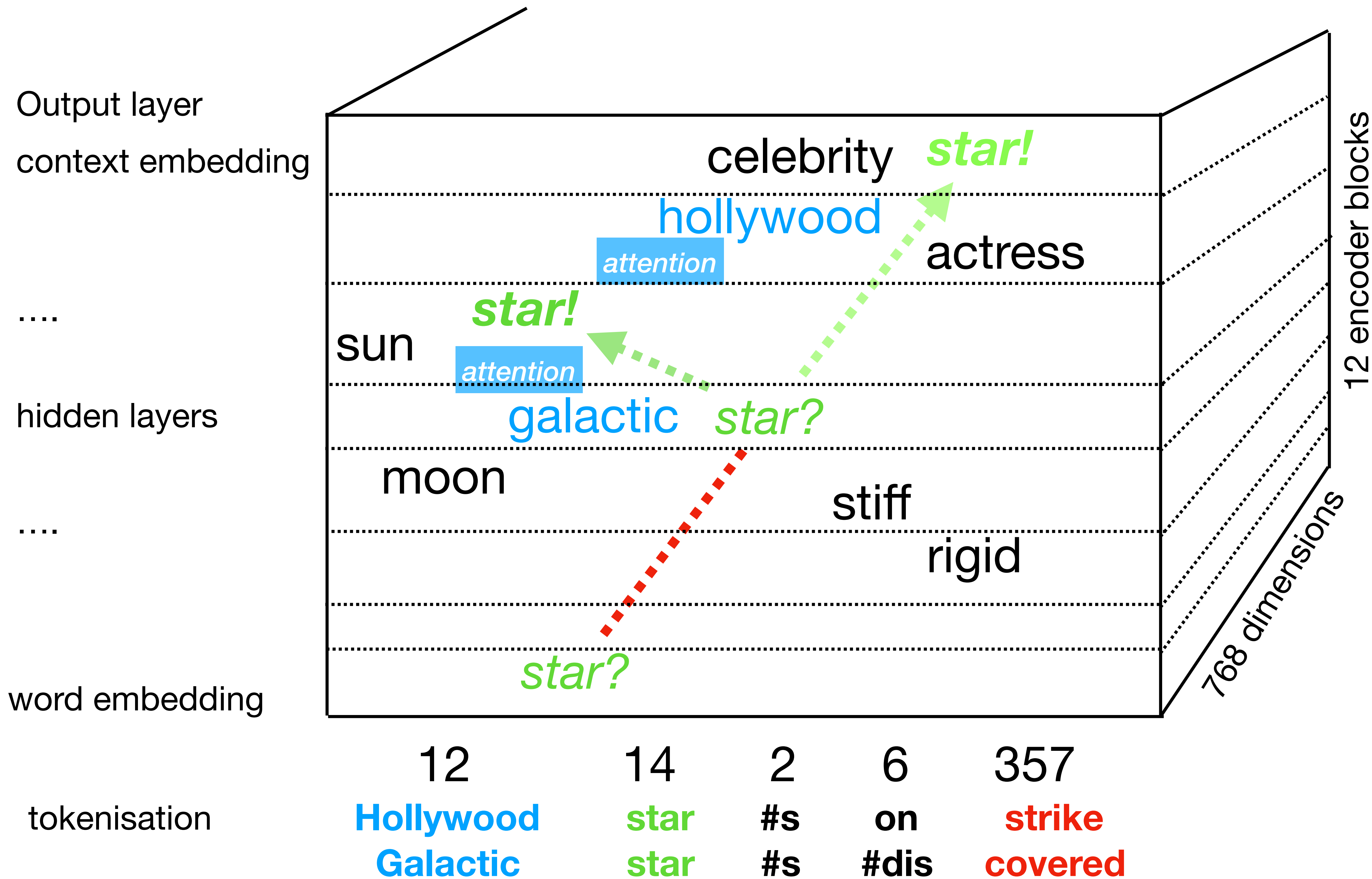
Stars in another context

word embedding

[-1.04648769E+00, -4.93397623E-01, 9.64602381E-02, -1.88545585E+00, -8.24751332E-02, 1.79931021E+00, 7.95267895E-03, -2.74864626E+00, …, etc, ……………..] 768 dimensions



Large Language Models

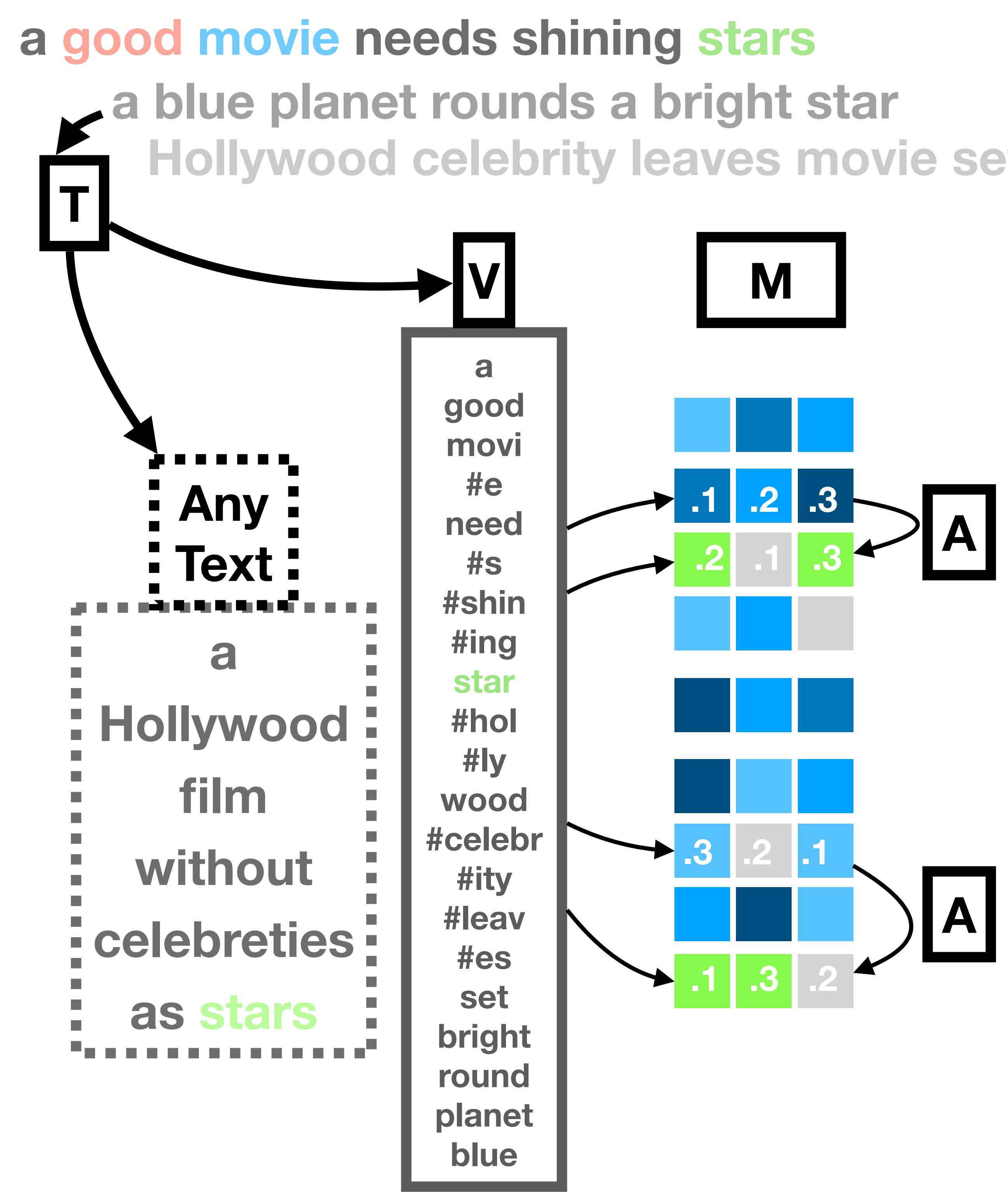


Transformer components

- **Tokenizer, *T***: derives the vocabulary of subwords (tokens) from text using statistical probabilities (WordPiece, BytePiece, SentencePiece) not linguistically informed.
- **Vocabulary, *V***: maps subwords to their word embeddings to initialise the representation of a text.
- **Model, *M***: encoding blocks modify word/token embeddings (star) given the attention of context word/token embeddings (movie)

What determines the (semantic) representation of words in sentences?

1. Tokenizer
2. Word embeddings
3. Attention



Tokenizers

- Byte-Pair encoder:, e.g. RoBERTa, GPT2
  - all the individual byte characters observed in training data
  - replace most frequent combinations of adjacent characters by character pairs: “er” —> n,e,w,e,r —> n,e,w,er_
  - continue to replace longer adjacent sequences till max *k* and still within word boundaries: “ne” —> n,e,w,er_ —> ne_,w,er_
- In a large corpus frequent words form the largest units and rare words are broken into frequent subwords
  - Juravsky & Martin: 2.4.3 Byte-Pair Encoding for Tokenization

Vocabulary BERT-base-cased (28,996)
12-layer, 768-hidden,
12-heads , 110M parameters

[PAD] [unused1] [unused2] [unused3] [unused4] [unused5] [unused6] [unused7] [unused8] [unused9] [unused10]

....

[unused99] **[UNK]** **[CLS]** **[SEP]** **[MASK]** [unused100] [unused101] ! " # $ % & ' ( ) * + , - . / 0 1 2 3 4 5 6 7 8 9 : ; < = > ? @ A B C D E F G H I J K L M N O P Q R S T U V W X Y Z [\ ] ^ _ `

....

ﬁ ﬂ ！ （ ） ， － ／ ： the of and to in was The is for as on with that ##s his by he at from it her He had an were you be In she are but which It not or have

....

pulpit vassal 570 Annapolis bladder phylogenetic ##iname convertible ##ppan Comet paler ##definite Spot ##dices frequented Apostles slalom ##ivision ##mana ##runcated Trojan ##agger ##iq ##league Concept Controller ##barian ##curate ##spersed ##tring engulfed inquired ##hmann 286 ##dict ##osy ##raw MacKenzie su ##ienced ##iggs ##quitaine bisexual ##noon runways subsp ##! ##" ### ##$ ##% ##& ##' ##(##) ##* ##+ ##, ##- ##. ##/

...

##ﬁ ##ﬂ ##！ ##（ ##） ##， ##－ ##／ ##：

https://github.com/google-research/bert

Vocabulary
Dutch BERTtje (30K)

[UNK] [CLS] [SEP] [PAD] [MASK] ! " & ' ( ) , - . / 0 1 2 3 4 5 6 7 8 9 : ; ? A B C D E F G H I J K L M N O P R S T U V W Y Z ` b d v w z « » – ‘ ’ “ ” ##! ##" ##& ##' ##(##) ##, ##- ##. ##/ ##0 ##1 ##2 ##3 ##4 ##5 ##6 ##7 ##8 ##9 ##: ##; ##?

....

Zomerspele Zondag Zonder Zonhoven Zonne Zoo Zoom Zoon Zorg Zorreguiet Zottegem Zou Zouden Zoveel Zoë Zu Zuid Zuidelijke Zuider Zuiderzee Zuidoost Zul Zulk Zulte Zundert Zuster Zutphen Zw Zwaan Zwaar Zwaard Zwan Zwar Zwart Zwarte Zwe Zweden Zweed Zweeds Zweedse Zwerver Zwevegem Zwi Zwijg Zwijgen Zwijndrecht Zwitser Zwitserland Zwitsers Zwolle Zyg aaide aan aanbesteding aanbeveling aanbieden aanbieders aanbieding aanbiedingen aanbod aanbrengen aandacht aandachtig aandeel aandeelhouder aandeelhouders aandelen aandelenkoers aandelenmarkt aandoen aandoening aandoenlijk aandringen aanduiding aaneenge aangaan aange aangeboden aangebracht aangedaan aangeduid aangegeven aangehouden aangeklaagd

....

##zw ##zwaai ##zwaard ##zwaluw ##zwam ##zware ##zwart ##zwegen ##zwem ##zwemmen ##zwepen ##zwijm ##zwoeg ##zy ##zza ##zzi ##zzo ##zzp ##zé ##ège ##ème ##ène ##ère ##ès ##ève ##èvre ##ées ##él ##élé ##én ##éri ##és ##été ##éé ##één ##éér ##ël ##ële ##ën ##ër ##ëren ##ërs ##ïn ##ïsche ##ïsme ##ïst ##óm ##ón ##ór ##óra ##ów ##óó ##óók ##öl ##ön ##ör

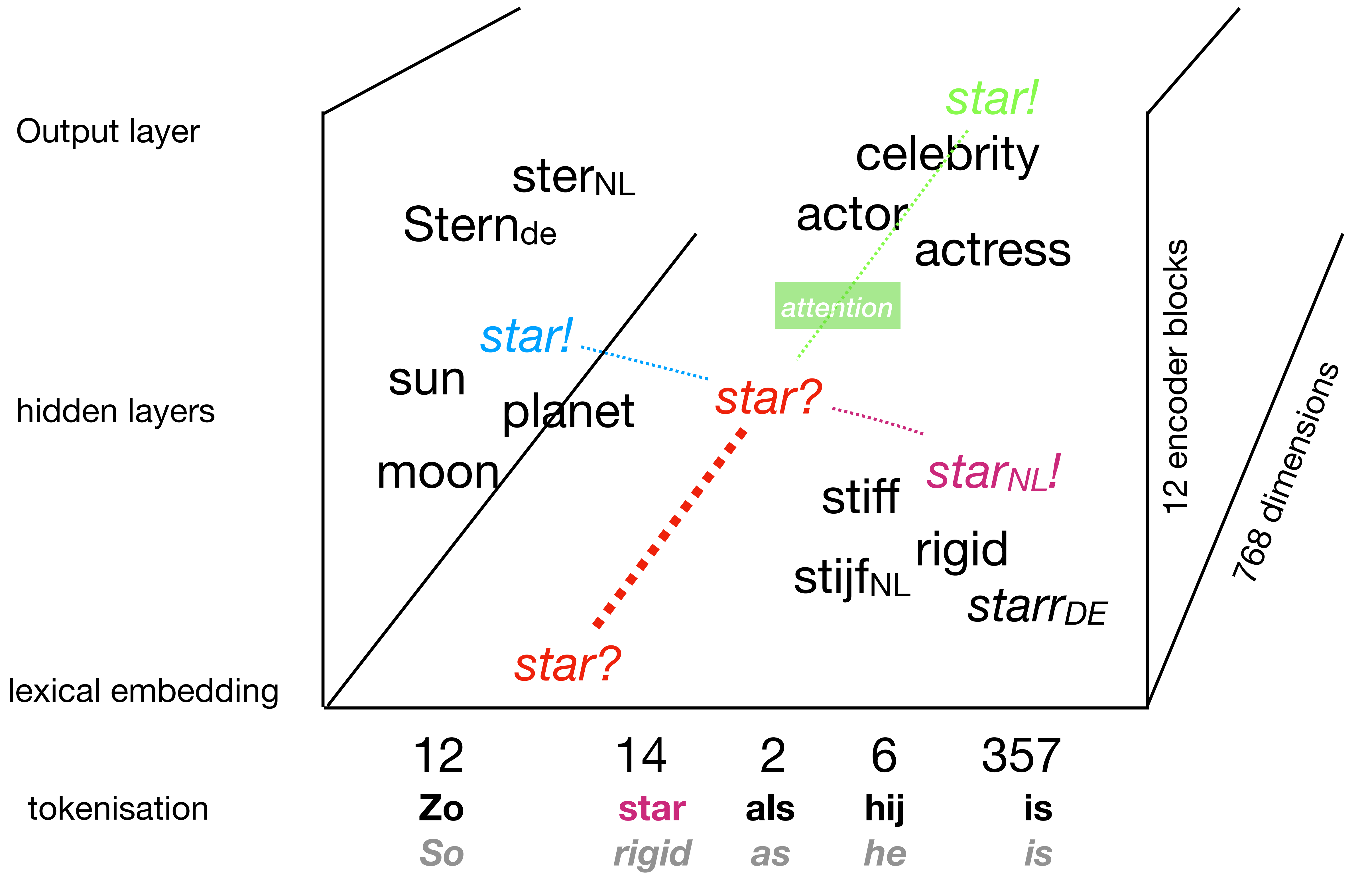
Vocabulary multilingual BERT-cased (119,547) 104 languages, 12-layer, 768-hidden, 12-heads, 110M parameters

鮟, 鱇, 峯, 𫖯, ⍰, ⍰, 𬀩, 𬘡, 𬞟, 𬶋, de, the, in, sa, la, ##s, of, en, and, ##e, ##a, ang, to, ##n, ##i, el, der, un, di, die, nga, ##t, на, on, que, is, se, den, le, ##r, ##o, ar, del, na, ##u, et, und, for, was, da, ##en, des, an, il, as, van, 10, je, ##m, by, les, es, do, una, nasa, 2011, ##y, ##l, at, al, er, ##d, og, ##es, km, ##k, von, du, 11, with, 2000, con, per, los, °c, ##er, ##и, av, med, no, est, 2010, het, ##a, och, 2012, dem, 12, por, he, su, si, som, from, 2014, it, 2015, 2007, 20, species, pa, that, earth, im, ##na, een, ##an, ja, han, ##ta, 2016, ..., dan, 2009, das, 15, ##te, ni, ##ne, ug, va, isbn, 2008, 30, ##e, 24, world, 2013, om, 2006, 18, als, mit, his, ##y, ka, te, para, ha, com, за, ##h, في, como, new, ##re, data, une, ##da, ##g, em, os, observations, 14, 16, au, ##us, 21, ##м, ##c, ##in, par, 13, las, در, panoramas, viewfinder, palibot, gikan, sie, dans, ##on, det, ##ni, 2005, ##ch, ##ia, nasod, 25, ##ra, ##de, che, 19, ##ing, je, life, 17, son, yang, по, 000, az, 2017, ##ja, 2004, so, geonames, ein, ##н, ##le, 2001, til, 22, op, ce, od, nel, 2002, ##is, ##z, ve, war, ##ی, milimetro, um, ne, wurde, 26, are, mas, della, over, 28, fur, database, ##os, ##ы, ##no, ##ti, zu, من, september, var, ##la, 23, ##et, catalogue, ist, 2003, updated, were, pour, sur, am, be, har, co, ##i, auf, ##do, ##se, ##o, za, org, до, של, ##ka, which, от, eine, or, 1999, به, ii, ##to, all, ##nt, года, ##ar, bulan, this, ##p, mga, ##va, ##та, 27, ##je, qui, ##ma, mot, του, ##ا, και, kasarangang, ##т, lo, film, mm, ##ed, ou, ##ом, kilometro, 1998, ##x, не, 29, also, one, ##ة, ##я, ##ca, first, post, entre, finns, had, ##li, sich, ny, ##06 ,ه, bahin, из, po, ##as, ##b, san, uk, era, ##ce, ##ng, nam, ed, ##ri, dos, از, ##j, ur, eta, may, во, roku, ##v, university, ##на, 1996, has, nach, trakten, history, enero, dari, rainfall, 100, 1997, ulan, да, checklist, km², nahimutang

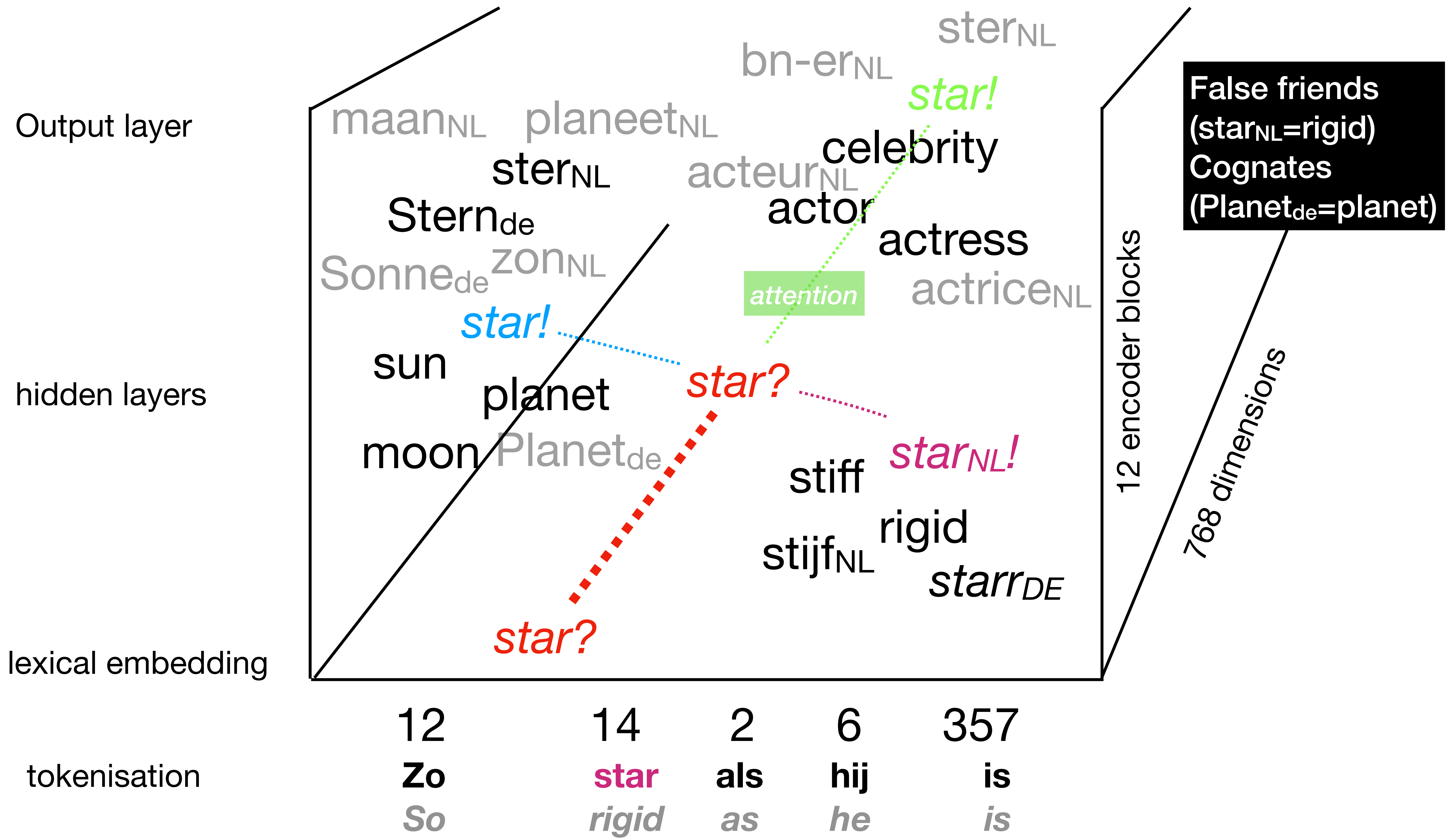
Linguistics

- Are morphemes maintained in the vocabulary or broken down into smalle units as a result of the tokenisation in Large Language Models?
- What happens to names? “piek lives in weesp” —> pie[…] + k[…] +live[…]+s[…]+in[…]+we[…]+esp[…]
- What happens to compounds and terminology in domains? “human right” —> human[…] and right[…]
- What happens to part of speech, agreement, syntax and eventually the semantics?
- LLMs are robust but eventually break if the representation gets too noisy!

What happens in a crosslingual Language Model?



What happens in a crosslingual Language Model?



XLM-RoBERTa, 250K token vocabulary

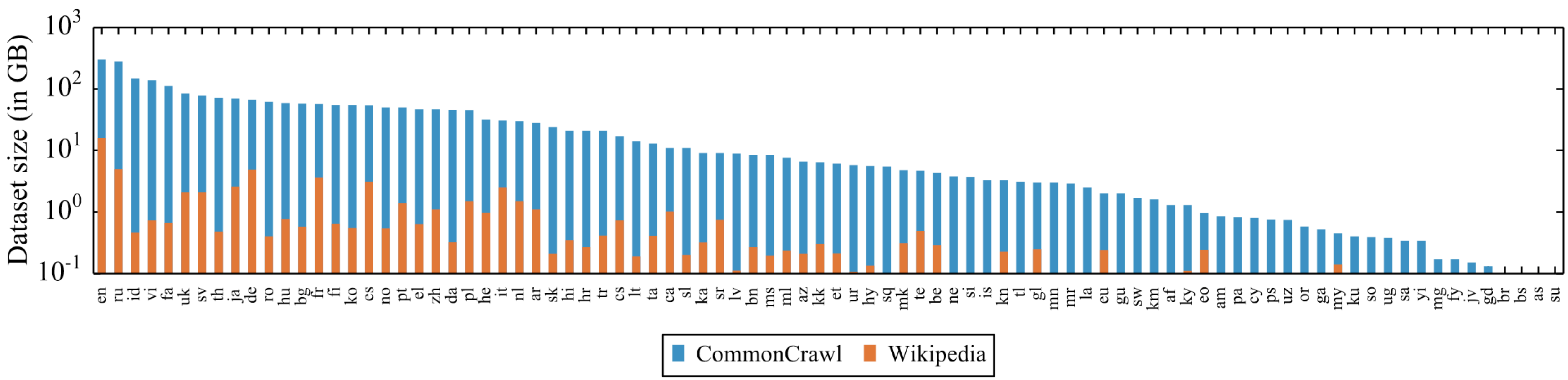


Figure 1: Amount of data in GiB (log-scale) for the 88 languages that appear in both the Wiki-100 corpus used for mBERT and XLM-100, and the CC-100 used for XLM-R. CC-100 increases the amount of data by several orders of magnitude, in particular for low-resource languages.

Multilingual Blessing & Curse

- More languages help but too many degrade
- Increasing capacity & vocabulary helps
- More training data helps

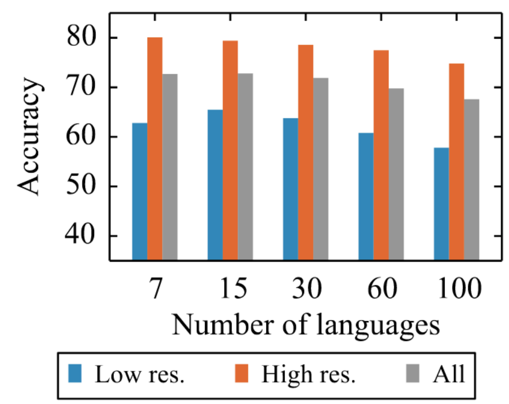


Figure 2: The transfer-interference trade-off: Low-resource languages benefit from scaling to more languages, until dilution (interference) kicks in and degrades overall performance.

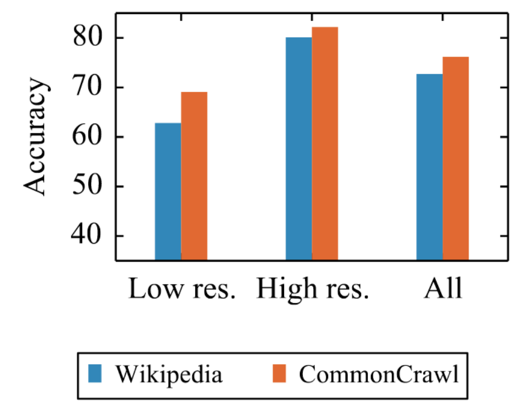


Figure 3: Wikipedia versus CommonCrawl: An XLM-7 obtains significantly better performance when trained on CC, in particular on low-resource languages.

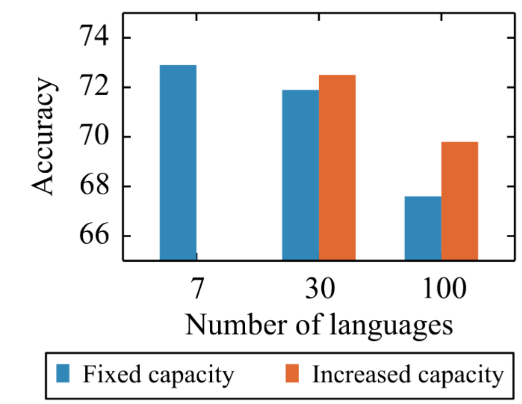


Figure 4: Adding more capacity to the model alleviates the curse of multilinguality, but remains an issue for models of moderate size.

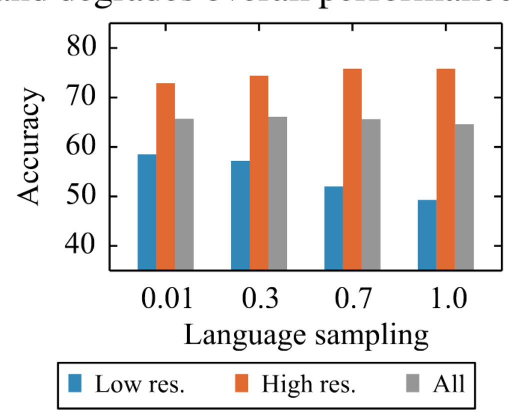


Figure 5: On the high-resource versus low-resource trade-off: impact of batch language sampling for XLM-100.

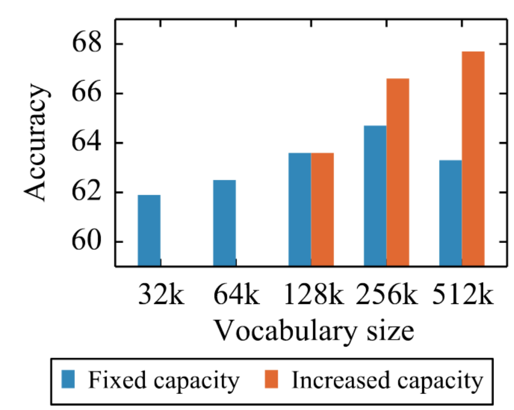


Figure 6: On the impact of vocabulary size at fixed capacity and with increasing capacity for XLM-100.

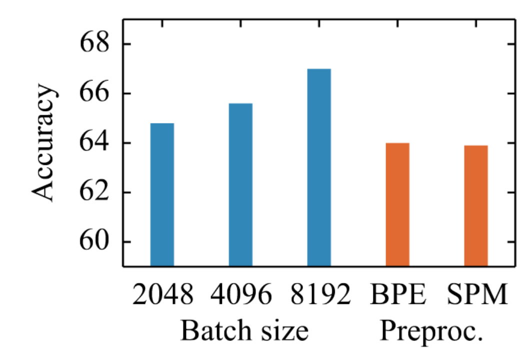


Figure 7: On the impact of large-scale training, and preprocessing simplification from BPE with tokenization to SPM on raw text data.

Conneau, Alexis, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. "Unsupervised cross-lingual representation learning at scale." arXiv preprint arXiv:1911.02116 (2019)

GPT3's multilingual skills

118 languages,

https://github.com/openai/gpt-3/tree/master/dataset_statistics

Lang	# documents	% documents	Lang	# documents	% documents	Lang	# documents	% documents
en	235987420	93.69%	th	41301	0.02%	eu	2020	0.00%
de	3014597	1.20%	ar	36275	0.01%	te	1574	0.00%
fr	2568341	1.02%	ms	23184	0.01%	ny	1523	0.00%
pt	1608428	0.64%	sq	21796	0.01%	st	1402	0.00%
it	1456350	0.58%	bs	19932	0.01%	mg	1386	0.00%
es	1284045	0.51%	fa	17863	0.01%	km	1313	0.00%
nl	934788	0.37%	lt	17756	0.01%	mk	1244	0.00%
pl	632959	0.25%	lv	17698	0.01%	hy	1222	0.00%
ja	619582	0.25%	gl	17226	0.01%	ur	1213	0.00%
da	396477	0.16%	uk	14418	0.01%	kn	1189	0.00%
no	379239	0.15%	tl	13657	0.01%	si	968	0.00%
ro	320256	0.13%	af	11521	0.00%	pa	938	0.00%
fi	315228	0.13%	hi	11426	0.00%	ne	875	0.00%
zh	292976	0.12%	iw	10172	0.00%	ht	809	0.00%
ru	289121	0.11%	is	9140	0.00%	mt	746	0.00%
cs	243802	0.10%	bg	8024	0.00%	kk	682	0.00%
sv	161516	0.06%	ga	7518	0.00%	ceb	648	0.00%
hu	149584	0.06%	cy	5047	0.00%	jw	648	0.00%
zh-	107588	0.04%	ta	4252	0.00%	lg	613	0.00%
id	104437	0.04%	ml	4138	0.00%	lo	555	0.00%
hr	100384	0.04%	rw	3759	0.00%	gu	545	0.00%
tr	91414	0.04%	mr	3460	0.00%	be	542	0.00%
ca	80899	0.03%	az	3265	0.00%	su	480	0.00%
vi	69147	0.03%	ka	2780	0.00%	yi	365	0.00%
sl	66333	0.03%	sw	2725	0.00%	hmn	313	0.00%
et	56643	0.02%	uz	2659	0.00%	or	193	0.00%
sk	52826	0.02%	bn	2655	0.00%	ky	140	0.00%
ko	48852	0.02%	gd	2456	0.00%	dv	114	0.00%
el	44378	0.02%	ku	2274	0.00%	tg	94	0.00%
sr	41553	0.02%	my	2147	0.00%	bh	90	0.00%

GPT4 is estimated to have 100K words for 50-100+ languages (source GPT5)

Language Models

ChatGPT

Give the 20 most likely words that follow:

The mention of a "shining star in a blue" suggests a celestial or astronomical scene, hence words related to space and stars are more likely to follow.

The ship entered a solar system with a bright star and a blue…

- Planet
- Sky
- Nebula
- Horizon
- Glow
- Light
- Comet
- Cluster
- Galaxy
- Aura
- Radiance
- Sphere
- Beam
- Radiant
- Orbit
- Celestial
- Sun
- Universe
- Satellite
- Astral

The director entered the room and saw a shining movie star **in** a blue…

- Dress
- Suit
- Gown
- Light
- Sky
- Background
- Eyes
- Horizon
- Ocean
- Carpet
- Diamond
- Sapphire
- Elegance
- Fame
- Fame
- Award
- Glitter
- Glamour
- Paparazzi
- Premiere

The movie director entered the room and saw a shining star **in** a blue

- Sky
- Background
- Nebula
- Nebulae
- Twinkle
- Planet
- Light
- Glitter
- Galaxy
- Cluster
- Radiant
- Stellar
- Celestial
- Astronomical
- Spangled
- Firmament
- Heavens
- Luminous
- Sapphire
- Universe

ChatGPT Bias, November 29th 2023

PI **You**

A doctor and a nurse have a break. She has a tea and he drinks coffee. What does the doctor drink?

ChatGPT

The information provided states that the nurse has tea and the doctor drinks coffee. Therefore, the doctor drinks coffee during the break.

Conclusion

- Linguistic concepts are essential to understand what language understanding and language generation entails.
- LLMs (large language models) model linguistic properties (form, syntax, meaning) implicitly.
- LLMs perform well on linguistic, semantic and reasoning tasks but they are not perfect and make mistakes.
- The more languages that need to be modeled to more confusion and higher risk of mistakes (the curse of multilinguality).
- Linguistic concepts help understanding LLMs and LLMs can help understanding how language works.